DATA 505 (Statistics Using R): Final Exam

Part 2: Open Computer - SOLUTION

Name: ____

2025-12-09

Instructions

Welcome to Part 2 of the exam!

You now have access to your computer and any online resources that are "not alive" (i.e., you may freely use online resources, but you may not communicate with any other person). You have approximately 45 minutes to complete this part.

Fill in the code chunks below and provide written answers where requested. Submit this completed `.qmd` file on Moodle along with the rendered PDF.

Problem 1: Two-Sample t-test (8 points)

We will analyze the `ToothGrowth` dataset, which contains data from an experiment on the effect of vitamin C on tooth growth in guinea pigs. The response is the length of odontoblasts (cells responsible for tooth growth) and one of the predictors is the delivery method (`supp`): orange juice (OJ) or ascorbic acid (VC).

```r
# Load the data
data(ToothGrowth)
```

1 (2 points)

Use the `str()` function to examine the structure of the `ToothGrowth` dataset. How many observations are there?

```r
str(ToothGrowth)
```

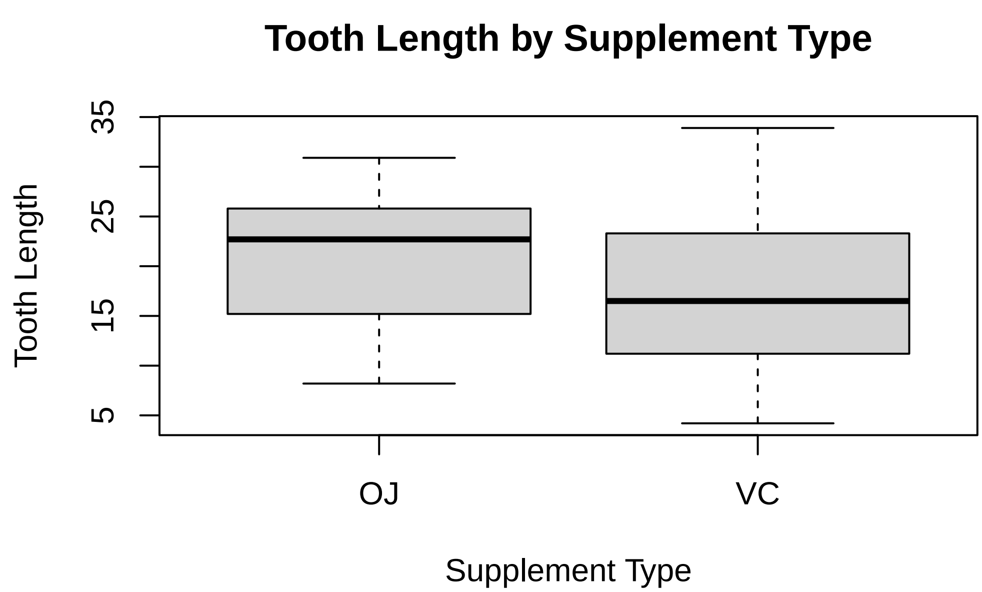
```
'data.frame':	60 obs. of  3 variables:
 $ len : num  4.2 11.5 7.3 5.8 6.4 10 11.2 11.2 5.2 7 ...
 $ supp: Factor w/ 2 levels "OJ","VC": 2 2 2 2 2 2 2 2 2 2 ...
 $ dose: num  0.5 0.5 0.5 0.5 0.5 0.5 0.5 0.5 0.5 0.5 ...
```

Number of observations: There are 60 observations in the dataset.

2 (3 points)

Create side-by-side boxplots comparing tooth length (`len`) between the two supplement types (`supp`). You may use base R or ggplot2. Include appropriate axis labels.

```r
# Base R solution
boxplot(len ~ supp, data = ToothGrowth,
        xlab = "Supplement Type",
        ylab = "Tooth Length",
        main = "Tooth Length by Supplement Type")
```



```r
# Alternative ggplot2 solution
# library(ggplot2)
# ggplot(ToothGrowth, aes(x = supp, y = len)) +
#   geom_boxplot() +
#   labs(x = "Supplement Type", y = "Tooth Length",
#        title = "Tooth Length by Supplement Type")
```

3 (3 points)

Perform a two-sample t-test to determine if there is a significant difference in mean tooth length between the two supplement types. Use $\alpha = 0.05$. State your conclusion in context.

```r
t_result <- t.test(len ~ supp, data = ToothGrowth)
t_result
```

```
	Welch Two Sample t-test

data:  len by supp
t = 1.9153, df = 55.309, p-value = 0.06063
alternative hypothesis: true difference in means between group OJ and group VC is not equal t
95 percent confidence interval:
 -0.1710156  7.5710156
sample estimates:
mean in group OJ mean in group VC 
        20.66333         16.96333 
```

Conclusion: With a p-value of 0.0606, which is greater than $\alpha = 0.05$, we fail to reject the null hypothesis. There is insufficient evidence to conclude that there is a significant difference in mean tooth length between the two supplement delivery methods (orange juice vs. ascorbic acid) at the 0.05 significance level.

Problem 2: Multiple Linear Regression (9 points)

We will analyze the `swiss` dataset, which contains fertility and socio-economic indicators for 47 French-speaking Swiss provinces around 1888.

```r
# Load the data
data(swiss)
```

1 (2 points)

Fit a simple linear regression model predicting `Fertility` from `Education` (percent of draftees with education beyond primary school). Store the model as `model1` and display the summary.

```r
model1 <- lm(Fertility ~ Education, data = swiss)
summary(model1)
```

```
Call:
lm(formula = Fertility ~ Education, data = swiss)

Residuals:
    Min      1Q  Median      3Q     Max 
-17.036  -6.711  -1.011   9.526  19.689 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  79.6101     2.1041  37.836  < 2e-16 ***
Education    -0.8624     0.1448  -5.954 3.66e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.446 on 45 degrees of freedom
Multiple R-squared:  0.4406,	Adjusted R-squared:  0.4282 
F-statistic: 35.45 on 1 and 45 DF,  p-value: 3.659e-07
```

2 (2 points)

Now fit a multiple regression model predicting `Fertility` from `Education`, `Agriculture` (percent of males in agriculture), and `Catholic` (percent Catholic). Store this as `model2` and display the summary.

```r
model2 <- lm(Fertility ~ Education + Agriculture + Catholic, data = swiss)
summary(model2)
```

```

Call:
lm(formula = Fertility ~ Education + Agriculture + Catholic,
    data = swiss)

Residuals:
    Min      1Q  Median      3Q     Max
-15.178  -6.548   1.379   5.822  14.840

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 86.22502    4.73472  18.211  < 2e-16 ***
Education   -1.07215    0.15580  -6.881 1.91e-08 ***
Agriculture -0.20304    0.07115  -2.854  0.00662 **
Catholic     0.14520    0.03015   4.817 1.84e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 7.728 on 43 degrees of freedom
Multiple R-squared:  0.6423,    Adjusted R-squared:  0.6173
F-statistic: 25.73 on 3 and 43 DF,  p-value: 1.089e-09
```

3 (2 points)

Compare the R-squared values of `model1` and `model2`. Does adding `Agriculture` and `Catholic` improve the model's explanatory power?

```r
r2_model1 <- summary(model1)$r.squared
r2_model2 <- summary(model2)$r.squared
r2_model1
```

```
[1] 0.4406156
```

```r
r2_model2
```

```
[1] 0.6422541
```

Answer: Yes, adding `Agriculture` and `Catholic` substantially improves the model's explanatory power. Model 1 has an R-squared of 0.4406 (explaining about 44.1% of the variance in fertility), while Model 2 has an R-squared of 0.6423 (explaining about 64.2% of the variance). This represents an improvement of about 20.2 percentage points in explained variance.

4 (3 points)

Use the `anova()` function to formally test whether `model2` is significantly better than `model1` at the $\alpha = 0.05$ level. What is your conclusion?

```r
anova_result <- anova(model1, model2)
anova_result
```

```
Analysis of Variance Table

Model 1: Fertility ~ Education
Model 2: Fertility ~ Education + Agriculture + Catholic
  Res.Df    RSS Df Sum of Sq      F  Pr(>F)
1     45 4015.2
2     43 2567.9  2    1447.3 12.118 6.7e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Conclusion: The ANOVA F-test yields a p-value of 6.7e-05 (approximately 7×10^{-5}), which is much less than $\alpha = 0.05$. We reject the null hypothesis and conclude that Model 2 (with Education, Agriculture, and Catholic as predictors) is significantly better than Model 1 (with only Education as a predictor). Adding Agriculture and Catholic as predictors significantly improves the model's ability to predict fertility rates.

Problem 3: Logistic Regression (8 points)

We will analyze the `birthwt` dataset from the `MASS` package, which contains data on risk factors associated with low infant birth weight.

```
# Load the MASS package and data
library(MASS)
data(birthwt)

# The variable 'low' is our binary outcome:
# low = 1 means birth weight < 2500g (low birth weight)
# low = 0 means birth weight >= 2500g (normal)

# The variable 'smoke' indicates whether the mother smoked during pregnancy:
# smoke = 1 means the mother smoked
# smoke = 0 means the mother did not smoke
```

1 (3 points)

Fit a logistic regression model predicting `low` (low birth weight) from `smoke` (smoking status during pregnancy). Store the model as `smoke_model` and display the summary.

```
smoke_model <- glm(low ~ smoke, data = birthwt, family = binomial)
model_summary <- summary(smoke_model)
model_summary
```

```
Call:
glm(formula = low ~ smoke, family = binomial, data = birthwt)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  -1.0871     0.2147  -5.062 4.14e-07 ***
smoke         0.7041     0.3196   2.203   0.0276 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 234.67  on 188  degrees of freedom
Residual deviance: 229.80  on 187  degrees of freedom
```

```
AIC: 233.8

Number of Fisher Scoring iterations: 4
```

2 (2 points)

Based on the model summary, is there a statistically significant relationship between smoking and low birth weight at the $\alpha = 0.05$ level? Justify your answer.

Answer: Yes, there is a statistically significant relationship between smoking and low birth weight. The p-value for the `smoke` coefficient is 0.0276, which is less than $\alpha = 0.05$. This means we can reject the null hypothesis that the coefficient for smoking is zero. The positive coefficient (0.7041) indicates that smoking during pregnancy is associated with an increased probability of low birth weight.

3 (3 points)

Use the fitted model to predict the probability of low birth weight for two mothers: one who smoked during pregnancy (`smoke = 1`) and one who did not (`smoke = 0`). Report both probabilities.

```r
# Create a data frame with the two scenarios
new_data <- data.frame(smoke = c(0, 1))

# Predict probabilities
predictions <- predict(smoke_model, newdata = new_data, type = "response")
predictions
```

```
        1         2 
0.2521739 0.4054054 
```

Probability of low birth weight (non-smoker): 0.252 (approximately 25.2%)

Probability of low birth weight (smoker): 0.405 (approximately 40.5%)

The model predicts that mothers who smoke during pregnancy have a substantially higher probability of delivering a low birth weight infant (40.5%) compared to non-smoking mothers (25.2%).